TraceBack

IIT Kanpur Lost & Found Portal

Deliverable 3 · Report 7

Defects Found and Fixed

Defect Register and Evidence for Deliverable 3

CS455 – Software Engineering · 2026-27

Member	Roll no.	Primary responsibility
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GitHub: https://github.com/cs455project/campus-lost-and-found

Jira: https://iitk-team-m5zcmuav.atlassian.net (project Group 4, key SCRUM)

Live system: https://16-176-186-149.sslip.io

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Summary

Deliverable 3 testing found 25 defects and weaknesses in TraceBack: 1 critical, 6 high, 12 medium and 6 low. All 25 are fixed, each fix is re-checked by the test, attack or audit that exposed it, and every suite passes on the final code. None of them broke sign-in, access control or the claim rules under normal use; they surfaced under load, under parallel requests, under attack, in failures, and in peer review.

TraceBack is the IIT Kanpur Lost & Found portal. This register is the defect history for Deliverable 3; IDs (D-01 to D-25) are used across the test reports. The team's Jira site is suspended, so the history is kept here and in the commit messages on the `d3/integration` branch.

1 Where the defects came from

Found by	Critical	High	Medium	Low	Total
Load test (Performance report)	1	1	1	2	5
Concurrency tests	–	3	3	1	7
Adversarial tests and library audits	–	1	3	2	6
Failure and recovery tests	–	1	3	–	4
Deliverable 2 peer reviews	–	–	2	1	3
All	**1**	**6**	**12**	**6**	**25**

2 Defect register

ID	Severity	Defect	Found by	Fix (commit on `d3/integration`)	Verified by	Status
D-01	Critical	The API locked up at 25 simultaneous users: 40 worker threads waited for 15 database connections while the requests holding them waited for a thread; nothing was answered until 30-second timeouts	Load test (ramp), CON-14	At most 30 requests in flight per process; pool sized to match (`423b3bf`)	Ramp test: 0 errors up to 100 users; CON-14	**Fixed**
D-02	High	Two guards could hand over the same item at once: 2 "collected" entries in 10 of 10 trials	CON-08	Item row, then claim row locked in every claim action (`c2b51ad`)	CON-08: 1 hand-over in 40 of 40	**Fixed**
D-03	High	One guard approving while another rejected the same claim recorded both decisions (2 of 10 trials)	CON-07	Same row locks (`c2b51ad`)	CON-07: 1 decision in 40 of 40	**Fixed**

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ID	Severity	Defect	Found by	Fix (commit on `d3/integration`)	Verified by	Status
D-04	High	A claimant withdrawing during hand-over left claim and item disagreeing (6 of 10 trials)	CON-09	Same row locks (`c2b51ad`)	CON-09: history consistent in 40 of 40	**Fixed**
D-05	Medium	Two guards could receive the same item at two offices: 2 receipts in 10 of 10 trials	CON-04	Item row locked in receive, transfer and dispose (`c2b51ad`)	CON-04: 1 receipt in 40 of 40	**Fixed**
D-06	Medium	10 parallel sign-in requests for one address sent up to 7 codes, skipping the 60-second wait	CON-01	Per-address advisory lock (`c2b51ad`)	CON-01: 1 code, 9 “please wait”	**Fixed**
D-07	Low	Sending the same claim 5 times at once gave 500 errors (30 of 50 responses)	CON-05	Duplicate caught by the database answered 409 (`c2b51ad`)	CON-05: 0 server errors	**Fixed**
D-08	Medium	A 136 KB image of 12,000 × 12,000 pixels needed 782 MB to decode	ADV-18	Images over 60 megapixels refused before decoding (`423b3bf`)	ADV-18 stopped (400); `test_performance_changes.py`	**Fixed**
D-09	High	List pages sent full-size 2.9 MB photos: 18–21 MB per page view, 99.97% of the bytes	Load test	640 px WebP thumbnails for lists (`423b3bf`)	0.7–1.0 MB per page view	**Fixed**
D-10	Medium	Photo and proof uploads ran on the shared event loop and blocked every other request	Load test (profiler)	Upload endpoints run in worker threads (`423b3bf`)	Ramp and campus tests	**Fixed**
D-11	High	Next.js 16.4.0 had 3 published advisories (2 critical)	ADV-24 (`npm audit`)	Upgraded to 16.4.5 (`0e64287`)	`npm audit`: 0 vulnerabilities	**Fixed**
D-12	Medium	Pillow 12.3.0, which decodes every upload, had 39 published advisories	ADV-25 (`pip-audit`)	Upgraded to 13.0.0 with fastembed 0.9.1 (`0e64287`)	`pip-audit`: none found	**Fixed**
D-13	Medium	Shared photos kept their GPS location (EXIF)	ADV-17	Metadata removed on upload, picture unchanged (`a541035`)	ADV-17 stopped; `test_hardening.py`	**Fixed**
D-14	Low	No security headers (nosniff, frame protection, Referrer-Policy); X-Powered-By sent	ADV-16, ADV-19	Headers in Next.js, FastAPI and Caddy (`a541035`)	ADV-16, ADV-19 stopped	**Fixed**
D-15	Low	API documentation public in production	ADV-21	`API_DOCS=false` on the server (`a541035`)	ADV-21: 404	**Fixed**

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ID	Severity	Defect	Found by	Fix (commit on `d3/integration`)	Verified by	Status
D-16	Medium	Seeded office text "PLACE-HOLDER: update with real directions" reached a student in an email (FND-0740)	D2 review of PR #19	No seeded directions; old text cleared; emails omit missing directions; `set-office` command (`a9aba2b`)	`test_hardening.py` (5 tests)	**Fixed**
D-17	Medium	Failed emails were never retried	D2 review of PR #20	Temporary SMTP failures retried twice, 2 s then 4 s (`a9aba2b`)	`test_hardening.py`; FR-08	**Fixed**
D-18	Low	The AI agent's limits were hard-coded constants	D2 review of PR #15	Moved to settings (`AGENT_*`) (`a9aba2b`)	`test_hardening.py`	**Fixed**
D-19	Medium	With the database down, requests got a bare 500 and some connections were dropped	FR-01	Clear 503 with Retry-After (`9fd794f`)	FR-01, FR-02	**Fixed**
D-20	Medium	A crash between saving a report and queueing its AI job left a report never analysed	FR-04	Report and job in one commit (`9fd794f`)	FR-04: 0 of 336 without a job	**Fixed**
D-21	High	An AI job interrupted by a killed worker failed for good; the page said "looking for matches" forever	FR-06	Interrupted jobs re-run, up to 3 tries (`9fd794f`, migration `7f3a2c9d41b8`)	FR-06: done 0.9 s after restart	**Fixed**
D-22	Medium	A sign-in code that could not be emailed blocked a retry for 60 s	FR-08	Unsent code removed (`9fd794f`)	FR-08: immediate retry 202	**Fixed**
D-23	Low	Extra database queries: 2 per signed-in request, 5 per photo, 21 for the staff claims page	Load test (query counter)	1, 2 and 9 queries (`423b3bf`)	`test_performance_changes.py`	**Fixed**
D-24	Low	The AI queue also ran inside the API, loading the AI model twice	Load test (configuration)	AI queue only in the worker container (`423b3bf`)	Deployment configuration	**Fixed**
D-25	Medium	Once uploads moved to threads, 6 parallel uploads could pass the 3-photo limit	CON-10 (during the D-10 fix)	Item row locked while photos are counted (`423b3bf`)	CON-10: 3 stored in 40 of 40	**Fixed**

3 Evidence for six significant defects

3.1 D-01 — the API locked up under load (critical)

- **Found:** in the ramp test, not one request succeeded at 25, 50 or 100 simultaneous users, and with normal pauses 99.4–100% failed from 50 to 400 users. The API's CPU fell to about 1%: it was waiting, not working.
- **Cause:** the profiler (py-spy) showed all 40 FastAPI worker threads waiting for one of 15 database connections, while the requests holding those connections waited for a thread for their next step.
- **Fix:** a small ASGI middleware (`core/capacity.py`) lets at most 30 requests work at once per process, with a pool connection for each; the rest wait without holding anything and get 503 after 20 s.
- **Verified:** 100 users flat out with 0 errors at 231–255 requests a second; 200 users browsing with 95% of requests under 0.85 s; CON-14 passes (Performance report).

3.2 D-03 — approve and reject recorded together (high)

- **Found:** CON-07 sent "approve" and "reject" for the same claim at the same instant; in 2 of 10 trials both were recorded, leaving the claim's history contradicting its status.
- **Cause:** each action read the claim, checked its status and wrote, without locking; two requests could both pass the check.
- **Fix:** every claim action locks the item row and then the claim row, always in that order (no deadlocks), and reads both again before checking the rules (`services/claims.py`, `_lock`). The same fix closed D-02, D-04 and D-05.
- **Verified:** CON-04 to CON-09 pass in 40 of 40 trials each, with no deadlock in the PostgreSQL log.

3.3 D-13 — GPS location in shared photos (medium)

- **Found:** ADV-17 uploaded a phone photo with GPS data; another student downloaded it with the location intact. Found items are photographed where they were found, often in hostels.
- **Fix:** `storage.strip_metadata` removes EXIF, XMP and IPTC data on upload. JPEGs are cleaned segment by segment without re-encoding, so the picture is byte-identical and only the rotation flag is kept; PNG and WebP are re-saved. A damaged photo that still carries metadata is refused rather than shared.
- **Verified:** ADV-17 stopped; 10 tests in `test_hardening.py` (GPS gone, pixels identical, rotation kept, damaged files refused).

3.4 D-21 — interrupted AI jobs never finished (high)

- **Found:** FR-06 killed the worker while it held an AI job. After the restart the job was marked "failed", the report never got matches, and its page showed "looking for matches" indefinitely.
- **Cause:** interrupted jobs were failed outright, judged by when they were queued rather than started.
- **Fix:** a `started_at` column (migration `7f3a2c9d41b8`); jobs running longer than 10 minutes are queued again, up to 3 tries, and after that the page says matching failed.
- **Verified:** FR-06 passes, finishing 0.9 s after the worker restarts; 5 tests in `test_failure_recovery.py`.

3.5 D-16 — placeholder text in a real email (medium)

- **Found:** in the Deliverable 2 review of PR #19, a teammate showed an owner email (FND-0740) that told the student to go to "PLACEHOLDER: update with real directions".

- **Fix:** the seed no longer invents directions and clears old placeholder text on every start; emails and pages leave out directions that aren't set; `python -m app.cli set-office` records the real ones.
- **Verified:** 5 tests in `test_hardening.py`, including an approval email with placeholder data in the database.

3.6 D-11 — a framework version with critical advisories (high)

- **Found:** `npm audit` (ADV-24) reported 3 advisories against Next.js 16.4.0, 2 of them critical (image optimisation, caching and Server Actions).
- **Fix:** upgraded to 16.4.5, a patch release; Pillow was upgraded at the same time (D-12).
- **Verified:** `npm audit` and `pip-audit` report nothing; the production build, the 15 system tests and the 23 attacks pass on the new versions.

4 Regression protection

Each fix is checked again on every run by the test, attack or audit that first exposed the problem: the concurrency and failure suites reproduce the races and crashes, the adversarial script repeats the attacks, and 35 pytest tests (`test_hardening.py`, `test_failure_recovery.py`) pin the security and recovery fixes. The full pytest suite (459 tests) runs in about 26 seconds.